

Task 1: Multi-Sensor Expansion Using FreeRTOS (Concurrent System Design)

Overview

This module demonstrates a real-time Smart Health Monitoring System using ESP32 and FreeRTOS. Multiple health and environmental parameters are monitored concurrently using independent FreeRTOS tasks. Queues, binary semaphores, mutexes, and the ESP32 Task Watchdog Timer ensure reliable communication, synchronization, and stability.

Objectives

Monitor ECG, Blood Pressure, Temperature, and Humidity; display readings on OLED; generate emergency alerts; demonstrate FreeRTOS multitasking and watchdog reliability.

Hardware

ESP32 DevKit V1, SSD1306 OLED, DHT22, ECG Sensor (simulated), Blood Pressure Sensor (simulated), LED, Buzzer.

Software

Arduino IDE, ESP32 Arduino Core, FreeRTOS, Adafruit SSD1306, Adafruit GFX, DHT Library, Wire Library.

FreeRTOS Tasks

Alert Handler (Priority 4), Read Health Vitals (3), Read Environment (2), OLED Display (2), MQTT Simulation (1).

Inter-Task Communication

Queue for sensor data, Binary Semaphore for alerts, Mutex for OLED access.

Watchdog Timer

ESP32 Task Watchdog monitors running tasks and helps recover from software hangs.

Features

Concurrent monitoring, OLED visualization, MQTT simulation, priority scheduling, queues, semaphores, mutexes, watchdog integration.

Expected Output

Continuous sensor monitoring, OLED updates, simulated MQTT messages, emergency alerts, stable multitasking.

Author

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